

PhonePe Transaction Insights

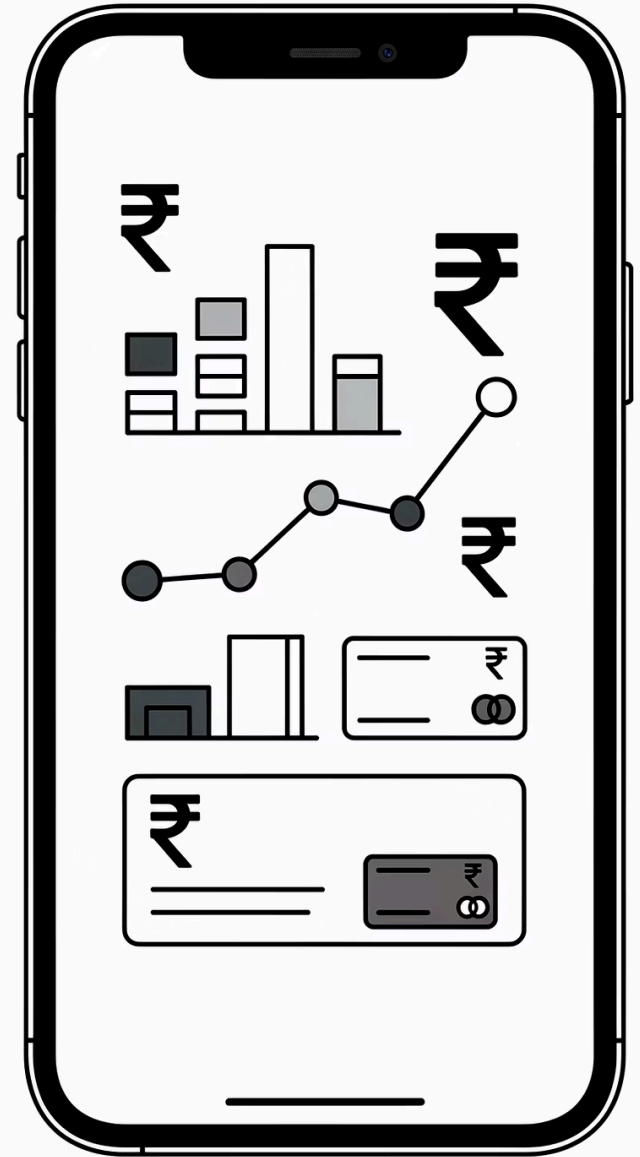
End-to-End Data Analysis using SQL, Python & Streamlit

Understanding transaction growth, user adoption, and insurance trends across India

Prepared by: Sai

Project Type: Data Analytics & Visualization

Year: 2026



Why this matters

Problem Statement

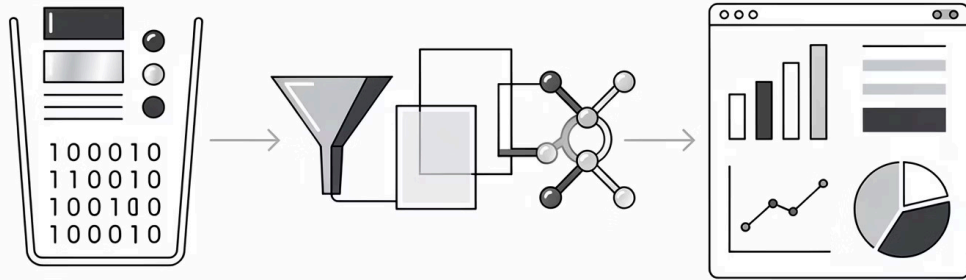
What problem are we solving?

PhonePe operates at massive scale across India, but raw transaction data alone does not directly answer:

- How fast are digital transactions growing?
- Which states contribute most to revenue?
- How are users adopting the platform?
- Is insurance becoming a meaningful growth driver?

Goal: Convert large-scale transaction data into **clear, actionable insights** for business and strategy teams.

Project Overview



What this project does

- Analyzes PhonePe Pulse transaction data across multiple years
- Converts raw data into structured analytical datasets
- Builds visual insights using Python and Streamlit
- Focuses on **growth, adoption, and diversification**

Core Domains Covered

- Transactions
- User growth
- Regional performance
- Insurance adoption

Business Objectives

Analyze year-over-year transaction growth

Identify top-performing states and regions

Understand user adoption trends

Evaluate insurance transaction growth

Provide insights usable for strategic decision-making

This project is designed to support:

- Business leadership
- Product strategy teams
- Market expansion planning

Data & Methodology

Data Source

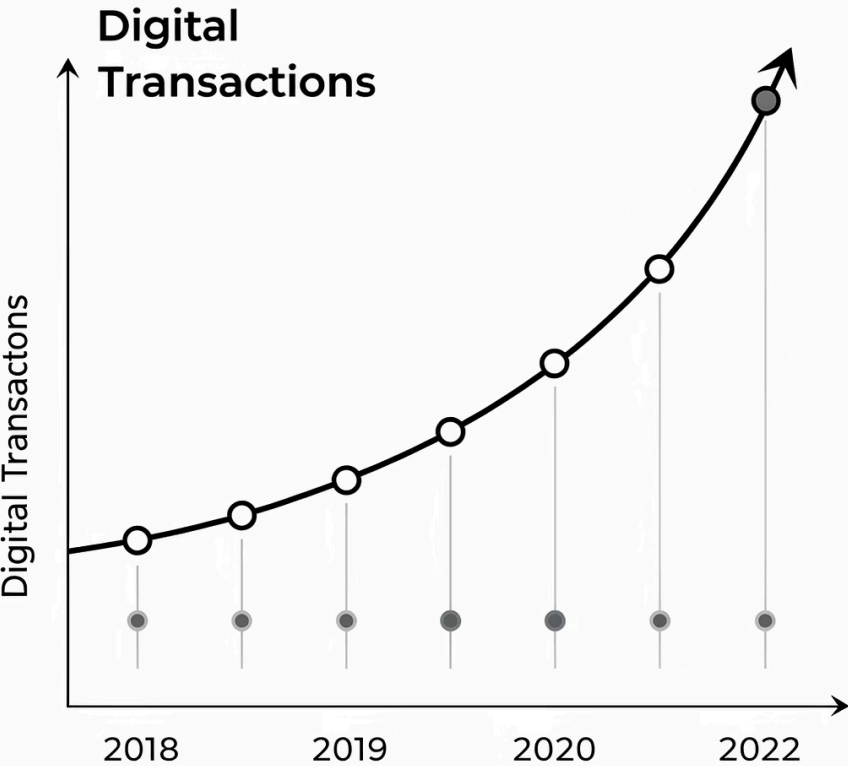
PhonePe Pulse (public aggregated datasets)

Approach

1. Data extraction and structuring
2. SQL-based aggregation and validation
3. Python-based analysis and visualization
4. Static insight generation
5. Dashboard presentation using Streamlit

Design Choice

Precomputed insights via DB queries → Faster performance → Better reliability → Cleaner storytelling



Yearly Transaction Growth

Insight

- Transactions show **consistent year-on-year growth**
- Strong acceleration observed after 2020
- Indicates rapid adoption of digital payments post-pandemic

Business Interpretation

- Digital payments have transitioned from optional to essential
- PhonePe is positioned as a core payment infrastructure

A stylized black-and-white outline map of India is shown. Inside the map, several icons are placed to represent different economic and industrial sectors. In the northernmost region, there is a bar chart with four bars of varying heights. In the north-central region, there are two stacks of coins. In the west-central region, there is another bar chart with four bars. In the central region, there is a single coin icon. In the south-central region, there is a factory building with two smokestacks. In the southernmost region, there is a stack of coins. In the easternmost region, there is a factory building with two smokestacks. The map is divided into several regions by thin lines, and the overall shape of India is clearly defined by a thick black outline.

- A small number of states contribute a disproportionate share of total transaction value
- Maharashtra, Karnataka, Telangana lead consistently

- High economic activity and digital maturity in these regions
- Opportunity to replicate success models in underpenetrated states

Dominant Transaction Types



Peer-to-peer payments

Dominate transaction volume



Merchant payments

High volume daily use



Financial services

Smaller but growing share

Insight

Peer-to-peer and merchant payments dominate transaction volume. Financial services and insurance form a smaller but growing share.

Business Interpretation

- PhonePe is primarily a daily-use platform
- Strong foundation for cross-selling financial products

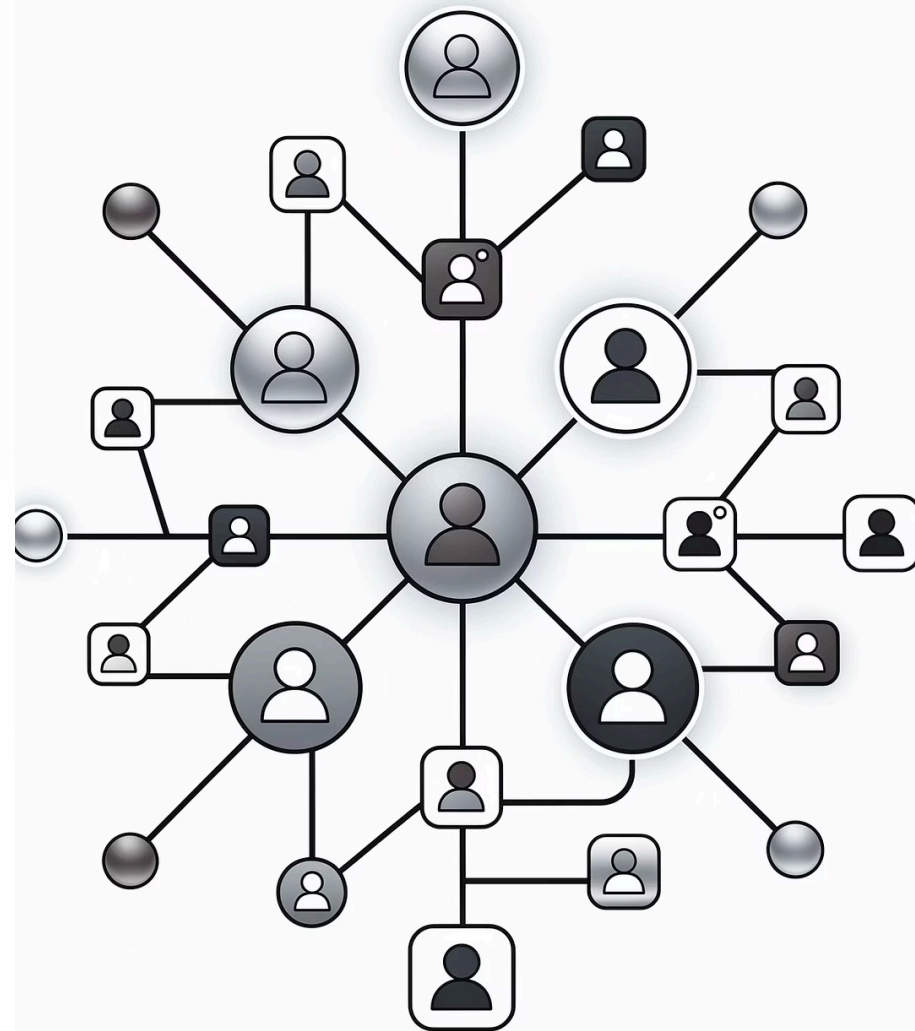
User Adoption Trends

Insight

- User registrations increase steadily every year
- No significant churn observed in the dataset

Business Interpretation

- ☐ Strong network effects
- ☐ High platform stickiness
- ☐ Trust in the ecosystem is increasing



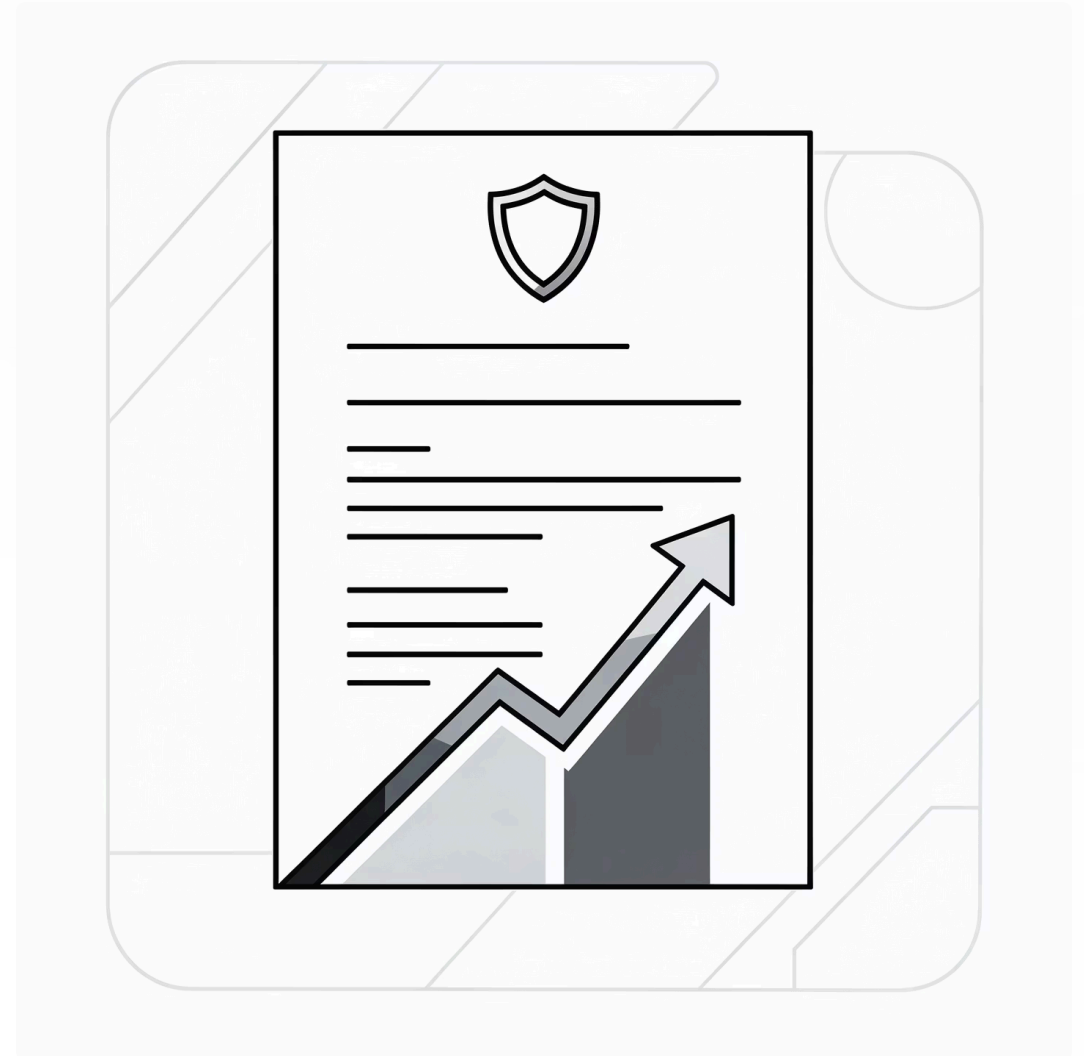
Insurance Transaction Growth

Insight

- Insurance transactions show noticeable growth in recent years
- Growth rate higher than overall platform average

Business Interpretation

- Insurance is emerging as a strategic diversification pillar
- Long-term revenue stability opportunity beyond payments



Dashboard Overview

What the Streamlit dashboard provides

Centralized
view of all key
insights

Static
visualizations
for
performance
efficiency

Clean layout
designed for
executive
consumption

Usage

- Analysts: quick insight access
- Leaders: trend-based decisions
- Demonstrations & reviews



Business Use Cases

How organizations can use these insights



Market expansion planning



Regional performance
benchmarking



Product adoption analysis



Insurance strategy development



Digital economy policy insights

Key Recommendations

Strategic Recommendations

01

Strengthen presence in high-performing states

02

Invest in user growth in underpenetrated regions

03

Expand insurance offerings aggressively

04

Use transaction behavior to personalize products

05

Monitor post-pandemic digital behavior trends continuously

Limitations & Future Scope

Current Limitations

- Uses aggregated public data
- No user-level behavioral tracking
- Static snapshot in time

Future Enhancements

- Real-time data pipelines
- Predictive modeling
- Deeper segmentation
- Live database-backed dashboards

Tools & Technologies Used



Python

data processing & analysis



SQL

data aggregation



MySQL

structured storage



Matplotlib / Plotly

visualizations



Streamlit

dashboarding



GitHub

version control



Conclusion

Final Takeaway

This project demonstrates how raw transaction data can be transformed into:

Clear insights

**Strategic
recommendations**

Executive-ready dashboards

It highlights PhonePe's growth, resilience, and future opportunities in India's digital economy.

Thank You

PhonePe Transaction Insights

End-to-End Data Analytics Project

Prepared by: *Sai*